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Fwd: Your VLSID 2027 Submission (Number 119)

1 message

Sandeep Joshi <sandeep.joshi@pilani.bits-pilani.ac.in> Mon, Oct 12, 2026 at 8:59 PM
To: SRIJAN MUKHOPADHYAY <f20230334@pilani.bits-pilani.ac.in>, YASH AGRAWAL <f20230306@pilani.bits-pilani.ac.in>

---------- Forwarded message ---------
From: **Sandeep Joshi** <sandeep85joshi@gmail.com>
Date: Mon, Oct 12, 2026 at 8:57 PM
Subject: Fwd: Your VLSID 2027 Submission (Number 119)
To: Sandeep Joshi <sandeep.joshi@pilani.bits-pilani.ac.in>

---------- Forwarded message ---------
From: **Technical Program Chairs, VLSID 2027** <vlsid2027@softconf.com>
Date: Mon, Oct 12, 2026 at 8:56 PM
Subject: Your VLSID 2027 Submission (Number 119)
To: <sandeep85joshi@gmail.com>

Dear Sandeep Joshi:

I am sorry to inform you that the following submission was not selected by the program committee to appear at VLSID 2027:

Edge AI-Based FPGA Accelerator for Spectrum Classification and Anomaly Detection

The selection process was very competitive. Due to time and space limitations, we could only choose a small number of the submitted papers to appear on the program. Nonetheless, I still hope you can attend the conference.

I have enclosed the reviewer comments for your perusal.

If you have any additional questions, please feel free to get in touch.

Best Regards,
Technical Program Chairs, VLSID 2027
VLSID 2027

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VLSID 2027 Reviews for Submission #119
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Title: Edge AI-Based FPGA Accelerator for Spectrum Classification and Anomaly Detection
Authors: Yash Agrawal, Srijan Mukhopadhyay and Sandeep Joshi


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                          REVIEWER #1
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Reviewer's Scores
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                 Research Novelty (1-5): 2
                Technical Details (1-5): 3
    Presentation and Writing Style (1-5): 3
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OVERALL RECOMMENDATION (1-5): 3

Detailed Comments to Authors

The paper develops a compact eight-layer 1D ResNet accelerator for four-class RF signal classification and implements it as hand-written INT8 Verilog on a Zynq-7000 FPGA. It combines residual CNN processing, BN folding, symmetric post-training quantization, and on-chip memory for a small FPGA-based RF classifier.

The paper provides a path from a trained model to synthesizable RTL and FPGA validation. The compact ResNet, BN folding, and INT8 implementation are suitable for a resource-constrained FPGA. The authors provide useful implementation details, including resource utilization, power, frequency, FSM organization, and memory requirements. The 29K-parameter model and small weight footprint are helpful for edge deployment.

However, several key issues need to be addressed:

1. Incremental novelty

BN folding, INT8 quantization, GAP through a shift, residual blocks, MAC engines, BRAM weight storage, and FSM-controlled RTL are all established techniques.

The authors should clarify what hardware technique is genuinely new beyond combining these established components.

2. Class detection

How are the 24 RadioML modulation classes mapped to Normal, Jammer, Spoofer, and Interference?

Also, please share how modulation type can reliably identify these operational categories, particularly Jammer?

3. Inference latency

The accelerator loads 128 I/Q pairs but claims only 56 cycles for complete inference. How are the 128 samples loaded within this latency?

4. Comparison with state of the art

The proposed accelerator should be quantitatively compared with the prior FPGA-based RF anomaly detector by Moss et al., which also targets real-time FPGA anomaly detection.

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REVIEWER #2
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Reviewer's Scores

Research Novelty (1-5): 3
Technical Details (1-5): 4
Presentation and Writing Style (1-5): 4
OVERALL RECOMMENDATION (1-5): 2

Detailed Comments to Authors

The paper presents a fully synthesizable FPGA accelerator for real-time radio frequency anomaly detection at the sensor edge. The target application is modern RF spectrum monitoring, where abnormal or hostile signals such as jamming pulses, spoofing transmissions, and wide-band interference must be detected with very low latency. The proposed system implements an eight-layer one-dimensional Residual Convolutional Neural Network, or ResNet-1D, on a Xilinx Zynq-7000 SoC.

The central contribution is the deployment of a quantized neural network accelerator that operates without cloud connectivity or external memory. The model achieves 83.23% accuracy in FP32 and 83.18% accuracy after INT8 symmetric post-training quantization, showing only a 0.05 percentage-point accuracy loss. The reported hardware design uses Batch Normalization folding and INT8 quantization to fit all weights into 28.6 KB of BRAM. The RTL datapath runs at 72.2 MHz and achieves an inference latency of 0.78 Î¼s with a reported on-chip power consumption of 0.671 W and energy efficiency of 0.38 Î¼J per inference.

However, there are several drawbacks in the current submission. First, there is no algorithmic description of the proposed methodology. Second, there is no overall block/architecture of the proposed hardware. Third, comparisons with the recent state-of-the-art publications are missing.

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REVIEWER #3
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Reviewer's Scores

Research Novelty (1-5): 2
Technical Details (1-5): 2
Presentation and Writing Style (1-5): 3
OVERALL RECOMMENDATION (1-5): 2

Detailed Comments to Authors

This paper presents an FPGA based accelerator for real-time RF anomaly detection at the sensor edge, eliminating dependence on cloud connectivity and external memory. The system implements an eight-layer one-dimensional CNN. The trained model achieves a test accuracy of 83.23% using 32-bit floating-point (FP32) precision and 83.18% after 8-bit integer (INT8) symmetric post-training quantization. The proposed hardware is implemented on a Xilinx Zynq-7000 SoC.

Comments:

1. The novelty of the proposed work appears to be limited, as the techniques employed are well-established and commonly used in similar FPGA based accelerator architectures. The authors should clearly identify and substantiate the specific novel contributions of the work.

2. What is the rationale for selecting an operating frequency of 72.2 MHz? should provide justification for this particular frequency and explain whether it is determined by the FPGA implementation constraints or any specific architectural consideration.

3. The paper reports that the LOAD operation and inference can be completed within 56 clock cycles. This appears to be very aggressive and requires further clarification.

4. A detailed FPGA-SoC block diagram is not provided. Including a detailed block diagram showing the processor, accelerator, memory hierarchy, interconnects, and other major components, along with relevant implementation details, would provide better clarity on the overall system architecture and implementation.

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REVIEWER #4
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Reviewer's Scores

Research Novelty (1-5): 3
Technical Details (1-5): 4
Presentation and Writing Style (1-5): 4
OVERALL RECOMMENDATION (1-5): 3

Detailed Comments to Authors

Summary: The paper presents a fully synthesizable FPGA accelerator on a Xilinx Zynq-7000 SoC for real-time RF anomaly detection at the sensor edge, implementing an 8-layer ResNet-1D compressed via INT8 quantization and BatchNorm folding to 28.6 KB of on-chip BRAM. The design achieves 0.78 Î¼s inference latency at 0.671 W power (0.38 Î¼J/inference).

Strengths: Effective model compression with minimal accuracy degradation (0.05 pp). Edge-oriented architecture with no external memory dependency. Adequate background, logical information flow, and

sufficient reproducibility detail.

Concerns:
(1) The four-class operational taxonomy (Normal/Jammer/Spoofer/Interference) and its mapping from RadioML 2018.01A's 24 classes is not clearly described.
(2). 83.23% accuracy is low for the stated mission For a security-critical task (countering jammers/spoofers in microseconds), a ~17% error rate is substantial.
(3) The GPU baseline comparison is methodologically unfair â€" comparing FPGA Datapath latency against batch-1 GPU end-to-end latency inflates the claimed advantage; fairer baselines (on-chip ARM cores, MCU-class processors, or comparable FPGA accelerators) should be used.
(4) The 72.2 MHz clock frequency is unusually low for the Zynq-7000 platform, suggesting potential timing closure issues or a serialized Datapath with limited parallelism.

Recommendation: Major revision â€" address the taxonomy description, accuracy justification, baseline fairness, and clock frequency clarification.

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VLSID 2027 - https://softconf.com/vlsi/vlsid2027

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